

Proactive blockage detection in sewer pipes using paired acoustic sensors: An experimental study

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ABSTRACT

Sewer overflows due to blockages are significant challenges for water utilities worldwide, as these events pose substantial risks of disruption and contamination to society, the economy, and the environment. Accurately detecting blockages in sewer systems is essential to mitigate such incidents. This paper investigates the use of paired acoustic sensors in combination with a paired-impulse response function (paired-IRF) technique to detect, locate, and estimate sewer blockages, helping to prevent sewer overflow risks. Laboratory tests were conducted on both clean and blocked pipes with varying blockage types and severities. The acoustic responses of these pipes were analysed to distinguish between unobstructed and obstructed conditions. The paired IRF extracted was then applied to identify blockage locations and estimate blockage sizes. The presence of paired spikes in the paired-IRF trace serves as clear evidence of blockages within the tested sewer pipe, including those caused by tree root and solid materials. The technique accurately identified both the location and size of blockages, showing strong agreement with actual conditions observed during experiments. These findings express that the acoustic-based paired-IRF approach is a reliable method for blockage detection in sewer systems, enabling water utilities to take informed and timely actions before issues lead to environmental or public health impacts.

1. Introduction

Sewer blockages remain a key challenge for water utilities worldwide, as they can lead to sewer overflows that have negative effects, including water pollution, the spread of diseases, property damage, and high maintenance and cleanup costs. Tree roots, sediments, oil buildup, and non-flushable items are the main causes of blockages in the sewer system. For example, Water UK reported that wet wipes accounted for 93% of the approximately 300,000 sewer blockages each year, costing the UK an estimated £100 million [1]. Similarly, over 3,500 blockages were built up from “unflushables” items (e.g., wet wipes, tissues, and paper towels) in the metropolitan Adelaide sewer network during the period of 2022–2023 [2]. SA Water (South Australia Water Corporation) spent more than two million dollars in 2024 to remove such materials from the sewer system to landfills [3].

Detecting sewer blockages at an early stage is difficult due to their unexpected occurrence and the complexity of wastewater systems [4,5]. Blockages can occur at various points in the system such as pipe segments, pipe connections and junctions, pump stations, and treatment facilities, making it difficult to monitor efficiently. Their severity can

vary from partial to complete, depending on the blockage source, the system layout, and changing flow conditions [6]. Additional factors like wastewater composition and climate variability further complicate both detection and response efforts [5]. These complexities require rapid and effective detection methods to mitigate potential impacts on the environment and infrastructure.

Effective sewer blockage monitoring depends on the timely and accurate identification of blockage locations. Traditional inspection methods such as closed-circuit television (CCTV) are accurate but time-consuming and expensive. Therefore, such methods are not effective for rapid inspections and large-scale network maintenance [7]. With the development of technology, real-time monitoring techniques using level sensors or flow sensors have been widely used in urban sewer systems for flood, water level, and flow monitoring [8–10]. However, these sensors present limitations when applying blockage detection. For instance, the performance of level sensors may be affected by the presence of foamy or vapor-producing liquids, and by complex junctions where multiple inflows and outflows complicate interpretation [8]. In addition, level sensors may have difficulty detecting blockages at an early stage, particularly in steep sewer systems where sensor coverage is

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limited [11]. Even when abnormal water levels are recorded at man-holes, indicating the presence of obstructions in the connected pipes, such measurements are insufficient to pinpoint the blockage location accurately.

In recent years, acoustic-based methods have been increasingly developed and applied for blockage detection in gas, water, and wastewater systems as these methods offer advantages such as low cost, simplicity, and the ability to capture the complex pipeline information about the internal pipe environment [9,12]. Many methods have been proposed based on acoustic reflectometry. This technique works by generating a broadband acoustic signal into the sewer pipe and analysing the reflected acoustic signals for anomaly detection. The effectiveness of this approach has been demonstrated in several studies. For instance, Monteiro et al. [13] developed a method using the reflected wave technique to identify blockages in gas pipelines. The study analysed the arrival time between measured signal reflections to estimate the blockage location and size. This method was validated through experimental tests, numerical simulations, and a full-scale onshore water-filled pipeline.

In the context of sewer systems, Yu et al. [14] developed analytical and empirical dispersion models for partially filled pipes, highlighting how wave-mode behaviour influences acoustic diagnostics in sewer pipe conditions. Building on the importance of wave propagation modelling, Yu et al. [15] demonstrated that an acoustic array with sparse-representation signal processing can localise blockages in a PVC pipe. Bin Ali et al. [4] also proposed an acoustic-based technique using the reflected acoustic signals to pinpoint blockages and other defect. The research focused on analysing the reflected acoustic intensity and frequency to assess pipe conditions. The method has been commercialized in a product named SewerBatt™. Another product, the Sewer Line Rapid Assessment Tool (SL-RAT™), has also developed and demonstrated by the United States Environmental Protection Agency (USEPA). Like SewerBatt™, SL-RAT™ transmits an acoustic signal through a pipe and estimates blockage severity based on signal attenuation. Additionally, Kelly et al. [16] combined the reflected wave technique with a time change detection algorithm to locate the blockage in the sewer pipe. Each of these acoustic-based detection methods has unique advantages and application conditions, but most work well in controlled environments with minimal lateral connections. Therefore, improvements are still needed to apply for complex systems with multiple lateral connections and background noise.

Beyond sewer networks, acoustic-based techniques have been widely applied for pipeline condition assessment, including leaks and blockages detection and localisation in water distribution systems [17]. Duan et al. [18] use the change in phase and power reflection ratio for identifying blockage shape in the pipeline. The impact of blockage sizes on the varying amplitude of the transmitted and reflected waves has been demonstrated by Abdullahi and Oyadiji [19] using the acoustic wave propagation technique. Moreover, impulse response function (IRF) methods were also applied by Lee et al. [20] for detecting leaks and blockages via identifying sharp pulses of all system reflections in a measured pressure response (i.e. one sensor). These sharp pulses enhance the accuracy of leak or blockage locations even when complex signals are injected into the system.

In acoustic-based blockage or leak detection methods, signal processing techniques such as cross-correlation and coherence analysis are commonly employed to interpret system responses to acoustic excitations. Cross-correlation estimates the time delay between signals measured at different points, supporting the identification of reflection

points that may indicate blockages [21–23]. Coherence analysis assesses the frequency-domain correlation between signals, helping to filter out random noise and highlight consistent structural anomalies [24]. These methods are widely used in commercial leak detection but have limitation when detecting multiple nearby leaks occurring at the same pipe [25].

A recent innovative method in acoustic analysis is the paired impulse response function (paired-IRF) method developed by Zeng et al. [26] for detecting leaks and other defects in pressurized water distribution systems. This approach extracts the impulse responses of a pipeline system using signals measured by two closely spaced sensors. Unlike single-trace methods mentioned in [20], the leak or blockage is represented by a pair of opposite-sign spikes on the paired-IRF trace, making it easier to distinguish from other objects in the system. Moreover, in the tests for pressurized water pipes, the method shows inherent robustness to environmental and operational noise, making it suitable for application in noisy and complex sewer environments. Although this method has advanced leak localization in pressurized water distribution systems, its extension to blockage detection in open-channel sewer networks has not been studied.

This study will investigate the use of paired acoustic sensors for detecting blockages in sewer networks. It was carried out through a series of experiments with PVC pipes under gravity flow and complex environmental noise conditions. The blockages of various sizes (low, medium, and large) were simulated to represent the real cases of sewer networks. The proposed method was assessed based on its robustness (i.e., how consistently it detects and locates blockages) and its sensitivity (i.e., how well it distinguishes between clean and blocked pipes, including estimating the size of a blockage from the spike magnitude). The experimental results are expected to highlight the practical value of acoustic techniques, especially the IRF approach using paired acoustic sensors, for reliably identifying, locating, and assessing the severity of blockages in gravity sewer systems, thereby contributing to effective environmental protection and management.

2. Methodology

The paired-IRF method is the primary approach employed in this study for detecting, locating, and estimating blockages in sewer pipes. To support the paired-IRF method in investigating blockage-induced changes in acoustic signals, frequency-domain analyses were utilized based on Power Spectral Density (PSD) and coherence functions.

2.1. Paired impulse response function (paired-IRF)

The paired-IRF method is based on wave propagation theory in the pipe that considers both the forward and backward pressure waves between two closely spaced microphones (M_1 and M_2). This formulation enables the deconvolution between two measured pressure signals to isolate the key impulse components associated with wave reflections from anomalies or defects. Detailed formulations of this method can be found in Zeng, et al. [26,27]. Following Zeng et al. [27], in this method, the transfer functions of the sections (the generator $-M_1$) and (M_1 - M_2) are denoted as H_1 and H_2 , and the impulse response function in the frequency domain at the left and right boundaries of M_1 are denoted as R_L and R_R , respectively. By considering both forward and backward wave interactions, the total pressures measured at M_1 and M_2 can be expressed in terms of these parameters. Substituting the directional pressure components into the wave equations gives the paired-IRF

relationship that links the two measured signals. The general expression of the paired IRF in the frequency domain for two acoustic traces $S_1(j\omega)$, $S_2(j\omega)$, j is the imaginary unit, and ω is the angular frequency, measured a short distance apart and considering all orders of wave reflections, is given by:

$$\frac{S_2(j\omega)}{S_1(j\omega)} = H_2 + \underbrace{\left(\frac{1}{H_2} - H_2\right)R_R}_{\text{order}=1} - \underbrace{\left(\frac{1}{H_2} - H_2\right)R_R^2}_{\text{order}=2} + \cdots + \underbrace{(-1)^{n-1}\left(\frac{1}{H_2} - H_2\right)R_R^n}_{\text{order}=n} + \cdots \quad (1)$$

where the first term H_2 represents the incident wave transmitted between the two transducers. The second term $\left(\frac{1}{H_2} - H_2\right)R_R$ corresponds to the first-order reflection, producing two spikes of opposite sign and separated by twice the one-way travel time between M_1 and M_2 . Higher-order terms correspond to higher-order reflections. As a result, a blockage induces a pair of spiked pulses with opposite directions and a fixed time interval equal to twice the propagation time from M_1 to M_2 . The detail of the derivation of Eq. (1) can be found in the [Appendix](#).

For practical applications subject to noise, higher-order terms that are on the order of the noise floor can be reasonably neglected. Therefore, a simplified form of Eq. (1) is adopted and used in this paper:

$$\frac{S_2(j\omega)}{S_1(j\omega)} = H_2 + \left(\frac{1}{H_2} - H_2\right)R_R \quad (2)$$

However, direct division in Eq. (2) can be numerically unstable once S_1 approaches zero, particularly at high frequencies. To address this, a least-squares deconvolution technique is applied in the time domain [\[28\]](#):

$$\mathbf{R}_t = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \quad (3)$$

where $\mathbf{X}$ is the lower triangular matrix built from the input signal in time domain $S_1(t)$; $\mathbf{y}$ is the vector of the pressure signal $S_2(t)$; $\mathbf{R}_t$ is the estimated impulse response vector in the time domain; $\mathbf{I}$ is the identity matrix; and λ is a positive regularization parameter, which can be adjusted to control the suppression of noise and stabilize the inversion. A larger λ imposes stronger smoothing on the estimated impulse response, reducing the influence of noise at the cost of potential resolution loss.

After deconvolution, the paired-IRF in the time domain consists of a superimposed pair of spiked reflections with opposite directions and a known time separation. Once the paired-IRF is extracted, the location of the blockage to the first sensor M_1 is then estimated based on the round-trip travel time τ of the excited wave from M_1 to the blockage and back to M_1 using:

$$x = \frac{a\tau}{2}, \quad (4)$$

where a is the wave speed in the pipe (approximately 343 m/s for sound travelling in air under laboratory conditions). τ can be obtained from the extracted paired-IRF trace by averaging the times of the pair of blockage-induced spiked pulses.

Once the location of a potential blockage is identified based on the time delay, the magnitude of the paired spikes in the paired-IRF trace is used to measure the ratio of the reflected wave to the incident wave r , which characterizes the impedance change at the blockage location. The reflection ratio at the location between the i th and $(i+1)$ th pipe sections can be also computed as [\[29,30\]](#)

$$r_{i,i+1} = \frac{B_{i+1} - B_i}{B_{i+1} + B_i} \quad (5)$$

where B_i and B_{i+1} are characteristic impedances at the i th and $(i+1)$ th pipe sections, respectively. The characteristic impedance is defined as: $B = a/gA$ where g is the gravitational acceleration; A is the cross-sectional area of the pipe section. When expressing the cross-sectional area as a ratio, A_i is normalized to 1 (i.e. 100%) as it represents the clean pipe

cross-sectional area. A_{i+1} is the ratio of the cross-sectional area at the blocked pipe section to the clean pipe area. By substituting and rearranging Eq. (5) within these components, the ratio of the blocked area to the clean pipe area (i.e., the blockage size) can be estimated directly from the reflection ratio, assuming constant wave speed and gravitational acceleration:

$$A_b = 1 - A_{i+1} = 1 - \frac{1 - r_{i,i+1}}{1 + r_{i,i+1}}. \quad (6)$$

To evaluate how excitation strength influences signal quality and the robustness of the paired-IRF method, a signal-to-noise ratio (SNR) analysis was conducted across five excitation-source volume settings (100%, 75%, 50%, 25%, and 12.5%) along with an additional baseline noise-only recording. The excitation level was adjusted directly on the broadband white-noise source by scaling its output amplitude prior to injection into the pipe system. For each signal, the power spectral density (PSD) was estimated using Welch's method. In addition, the root-mean-square (RMS) amplitude of each excitation signal was computed and compared with the RMS amplitude of the background noise-only dataset to determine the signal-to-noise ratio (SNR). The SNR was calculated from the ratio of RMS amplitudes using

$$\text{SNR} = 20 \log_{10} \left(\frac{\text{RMS}_{\text{signal}}}{\text{RMS}_{\text{noise}}} \right), \quad (7)$$

which provides a global measure of the excitation strength relative to background noise. For each test, the peak amplitude of the main excitation spike generated by the injected white-noise signal was extracted to represent the reference signal level. The blockage-induced reflection was then identified from the paired spike in the paired-IRF trace, and its average amplitude was computed across repeated measurements. This procedure provides a consistent and quantitative assessment of how excitation-source volume affects the relative strength of blockage reflections, enabling comparison of detectability under different acoustic input conditions.

2.2. Power spectral density (PSD) and coherence functions

In addition to the time-domain analysis, the PSD and coherence functions were employed to further evaluate acoustic signals measured in the tests. The PSD of the signals recorded at M_1 and M_2 was calculated to assess how consistently the spectral energy was distributed for each experimental condition. In this study, the PSD of the measured signals was calculated using the Welch method with a Fast Fourier Transform (FFT) and a Hann window. The signal had a length of 60 s and a sampling rate of 10 kHz. Each segment used in the analysis contained 1,024 data points, with 50% overlap between consecutive segments. By comparing PSDs of signals recorded changes in amplitude and spectral energy distribution could be observed, helping verify the reliability and consistency of data across different pipe condition.

Moreover, the coherence function between S_1 and S_2 signals was computed to quantify the linear relationship between signals at different

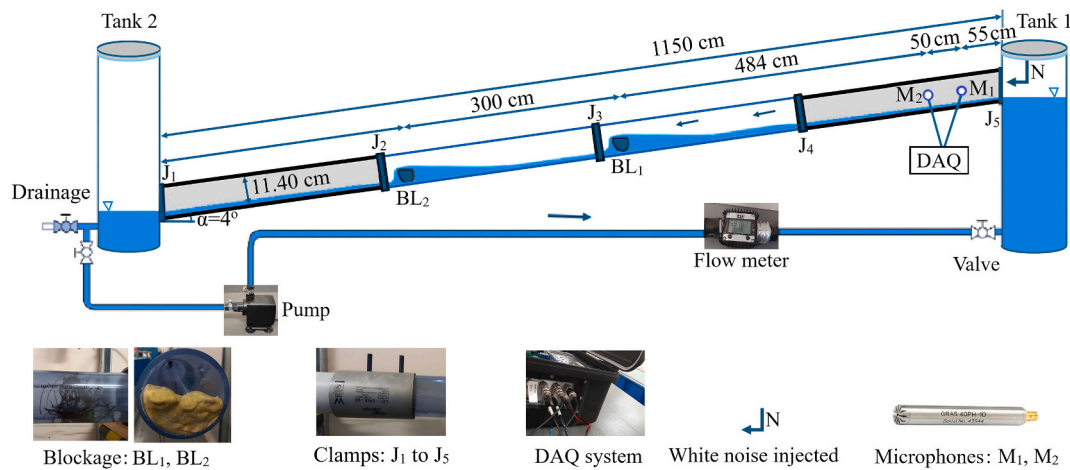


Fig. 1. The designed rig for sewer pipe in the laboratory.

frequencies. High coherence values indicate a strong correlation, suggesting that the signals are less disturbed by noise or external factors. In this study, PSD and coherence analyses provide supporting evidence for the reliability of the acoustic data, reinforcing confidence in the results obtained from the paired-IRF time-domain analysis. All signal processing, including Paired-IRF method, PSD, and coherence analysis, was performed using MATLAB.

3. Sewer pipe design in the laboratory

3.1. Experimental setup

To validate the ability of acoustic sensors in detecting sewer blockages, laboratory experiments have been conducted using a PVC sewer pipe setup at the Robin Hydraulics Laboratory, University of Adelaide. The experimental rig consisted of four pipe sections with an outer diameter of 114 mm and a wall thickness of 6.0 mm for each pipe. The total pipe length of the system was 11.5 m. The upstream and downstream sections were standard PVC pipes connected to two cylindrical tanks with an outer diameter of 225 mm. The two middle sections were made of clear PVC and connected via couplings (as shown in Fig. 1), allowing easy insertion and removal of blockages and enabling direct observation of hydraulic conditions during the experiments. The pipe was installed with a steep slope of 4° (i.e. 6.97%) to simulate a gravity sewer flow.

The rig was designed using a 25.4 mm hosepipe and a centrifugal pump (Model: RP2400, Manufacturer: REEFE) to circulate water between the two tanks. The pump has a maximum flow rate of 2300 L/h, a maximum head of 2.4 m, and a power rating of 40 W. During experiments, a flow rate of 15.2 L/min corresponding to a water depth of approximately 10 mm in the pipe was monitored using a digital flow meter (Model: K24, Manufacturer: PIUSI) with a measurement range of 7–120 L/min and an accuracy of $\pm 1\%$ of the reading. Flow rate adjustments were made via control valves installed in the system.

Two blockage types (tree root and solid blockages) were manually inserted into the pipe through joint J_2 or J_3 . The solid blockages were made from clay material in varying sizes. For each blockage test, a white noise signal was generated as an excitation signal using a mobile phone (N) placed inside the upstream tank (Tank 1). The acoustic reflections were collected by two GRAS 40PH free-field microphones positioned at M_1 and M_2 along the pipe. The distance between M_1 and M_2 was 0.5 m to ensure that the two main spikes in the paired-IRF trace are distinguishable and could be accurately grouped together, providing optimal paired-IRF estimation accuracy when using a broadband excitation source such as white noise. These microphones have a wide frequency range (10 Hz to 20 kHz), a sensitivity of 50 mV/Pa, and a dynamic range

of 15–134 dB SPL. The microphones were connected via RG174 cables to a National Instruments CompactRIO (cRIO)-based data acquisition (DAQ) system, equipped with a cRIO 9040 controller and a GPS time synchronization card. The DAQ system sampled acoustic signals at a rate of 10 kHz. Throughout the tests, the air temperature in the lab was approximately 20°C, corresponding to a sound speed in air of about 343 m/s. To mitigate the limitations of conducting experiments in a laboratory environment, which cannot fully replicate full-scale sewer conditions, all tests were under naturally occurring disturbance conditions present during the experimental period. These disturbances included real construction activities located approximately 20 m from the laboratory (e.g., hammering, drilling, cutting concrete), together with ambient noise from human activity (walking and talking), mechanical vibration, and pumping equipment. These noise sources were not artificially introduced but were passively recorded as part of the realistic laboratory environment.

3.2. Experimental procedure and tests

A structured experimental procedure was followed to minimise errors and ensure the accuracy and reliability of results. This involved setting up and calibrating the performance of the acoustic sensors and devices before conducting experiments under predefined flow and blockage conditions. For each experiment, acoustic data was recorded and checked for signal quality and re-collected if necessary. The process continued iteratively until all planned tests were completed, ensuring consistent and high-quality measurements across all experiments.

The experimental tests are designed to evaluate acoustic characteristics under different flow and blockage conditions within a PVC pipe. Each test was conducted for 60 s, ensuring consistent data collection across various scenarios. This study focuses on proactive blockage detection, identifying blockages at an early stage before they escalate into severe issues. Therefore, small and medium blockage cases were prioritised and tested multiple times to evaluate the sensitivity and robustness of the acoustic sensing technique for early-stage detection. A summary of the conditions for each laboratory experiment is shown in Table 1.

The first three tests focused on characterizing background noise under different conditions to establish a reference for signal understanding. Test 1 was conducted to examine the environmental noise inside the pipe when no water was flowing. The purpose was to establish a baseline noise level, helping to distinguish and evaluate the influence of other factors added to acoustic signals in subsequent tests. Test 2 was performed to record the environmental noise while water was flowing through the pipe. This test aimed to determine how flow influences background noise levels, which is important for understanding how

Table 1
Summary of laboratory experiment conditions.

Tests	Experimental conditions							
	Environmental noise	Flow rate	External excitation source (White noise)	Small solid blockage	Medium solid blockage	Large solid blockage	Tree root	Blockage location
1	✓							
2	✓	✓						
3	✓				✓			BL ₁
4	✓	✓	✓					
5	✓		✓		✓			BL ₁
6	✓	✓	✓	✓				BL ₁
7*	✓	✓	✓		✓			BL ₁
8	✓	✓	✓			✓		BL ₁
9	✓	✓	✓	✓	✓			BL ₂ , BL ₁
10	✓	✓	✓	✓			✓	BL ₂ , BL ₁
11**	✓	✓	✓	✓				BL ₁

* Conditions for Test 7 was used to run sensitivity analysis scenarios for signal-to-noise ratio with different excitation volume levels (100%, 75%, 50%, 25%, and 12.5%).

** Test 11 were conducted within sensors placed downstream of the pipe.

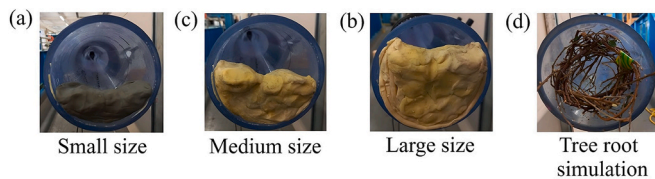


Fig. 2. Different blockage sizes and types used in blockage detection tests. The small and medium blockages had a thickness of 40 mm, while the large blockage had a thickness of 35 mm.

external sounds interact with moving fluid. Test 3 assessed the effect of a blockage in a dry pipe on the background noise.

Tests 4 and 5 investigated the impact of water flow and pipe blockage using actively injected white noise. In Test 4, a controlled white noise signal was generated while water was flowing, providing insights into how acoustic waves propagate in a flowing pipe. Test 5 examined the acoustic response when white noise was injected into a blocked pipe with no flow. This test was expected to find a clear indicator of blockage presence on the paired-IRF trace through reflected signals recorded without the fluid movement impacts.

To evaluate the ability of the paired-IRF method in detecting different levels of obstruction under flow conditions, Tests 6, 7, and 8 were conducted corresponding to three different blockage sizes (Fig. 2) located at BL₁. In Test 6, a small blockage (37% coverage, approximately one-third of the cross-sectional area of the pipe) was inserted, with white noise and flow present, to evaluate whether the paired-IRF method can

detect such partial obstructions under realistic conditions. Test 7 followed the same setup but used a medium-sized blockage (54% coverage) to investigate how increasing obstruction can be shown in the paired-IRF trace. Test 8 introduced a larger blockage with coverage of 82%, approximately four-fifths of the cross-sectional area of the pipe. Larger blockages are expected to generate stronger signal reflections (larger spike in the paired-IRF trace), making them more easily detectable.

Tests 9 and 10 were designed to explore the ability of the paired-IRF method in detecting multiple blockage events and different blockage types. To do that, two solid blockages were placed at locations BL₁ (medium size) and BL₂ (small size) in Test 9. Test 10 involved a tree root blockage (Fig. 2) at BL₁ and a small solid blockage at BL₂. Through these tests, the study aims to develop an effective acoustic-based method for detecting and locating real blockage scenarios in sewer systems, with a particular focus on how blockage material influences detection accuracy.

The impact of sensor placement on the paired-IRF results was also evaluated in Test 11, where both sensors were positioned near the downstream end of the pipe. In this test, a small blockage was placed at location BL₁, and the acoustic excitation source (white noise) was generated from the downstream tank (Tank 2).

4. Data analysis and results

4.1. Acoustic signal in different hydraulic and blockage conditions

Fig. 3 shows Power Spectral Density (PSD) curves for six different tests that were used to examine how flow, blockages, and white noise affect acoustic energy across various frequencies. The first three tests represent passive acoustic conditions and serve as baseline references. It can be seen that these background spectra are generally characterized by lower magnitudes. Test 2 (flow only) consistently shows the highest power levels across almost the entire frequency spectrum. This indicates that the fluid movement introduces additional background noise in the pipeline. In the dry pipe conditions, the PSD curves for Test 3 (blocked pipe, no flow) are relatively similar to Test 1 (empty pipe).

In contrast, once white noise is actively injected into the pipe, the PSD levels rise significantly across the entire frequency spectrum as seen in Tests 4, 5, and 6. This confirms that the injected signal improves signal strength. When white noise is injected into a blocked pipe without flow (Test 5), the PSD curve shows not only increased energy but also a change in curve shape compared to Test 3. In Test 6 (white noise, flow, and a blockage), the PSD retains high energy and a similar shape to Test 4 during most frequency bands except for variability in high frequencies.

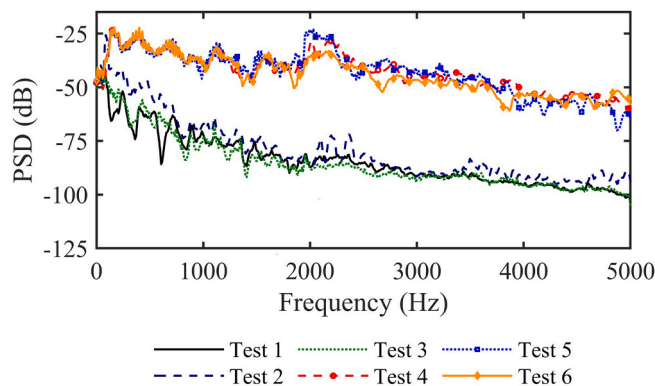


Fig. 3. PSD of signals recorded by sensor M₁ for different background conditions in sewer pipe. Test 1 – the empty pipe; Test 2 – flowing pipe; Test 3 – blocked dry pipe; Test 4 – white noise with flow; Test 5 – white noise with blockage only; and Test 6 – white noise with flow and blockage.

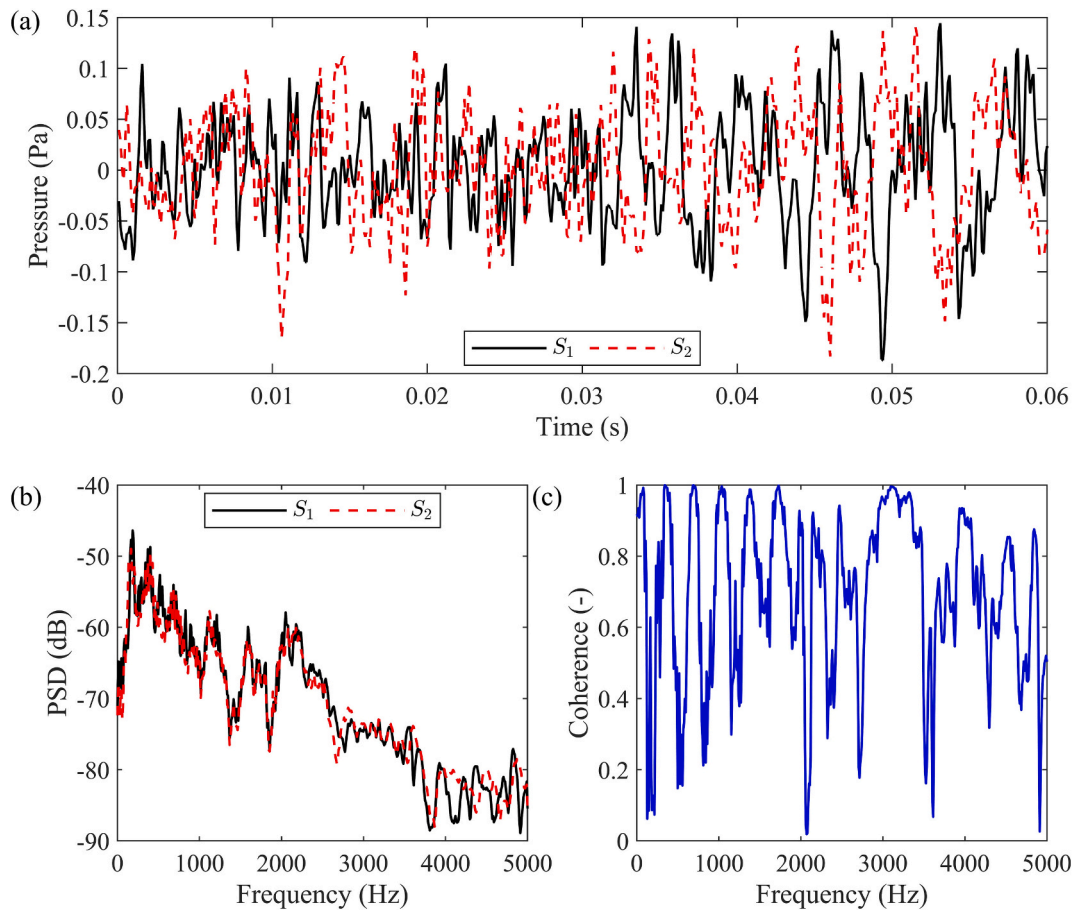


Fig. 4. Pressure signals, power spectral density (PSD), and coherence for Test 7 (medium blockage, 54% coverage). (a) Measured acoustic pressure at sensors M_1 and M_2 ; (b) PSD comparison between signals S_1 and S_2 ; (c) Coherence function between S_1 and S_2 .

4.2. Quality assessment of representative acoustic signals

Before extracting the paired-IRF traces, it is necessary to assess the quality of the measured signals. To achieve this, a set of representative signals measured in the case of blockage size 2 (Test 7) were assessed in the time domain and frequency domain (see Fig. 4). Fig. 4(a) shows a segment of the time series measured at M_1 and M_2 . Fig. 4(b) shows the PSD of the corresponding signals with a highly similar trend seen at two sensors, which is expected due to their close deployment.

The coherence function between the signals measured at M_1 and M_2 (Fig. 4(c)) shows that high coherence values appear over most of the frequency range. This demonstrates a strong linear relationship between the two signals. Periodic notches can be observed in the coherence function, which can be ascribed to the interference of the incident wave (propagating from M_1 to M_2) and the reflected wave (propagating from M_2 to M_1) by the blockage and Tank 2. Overall, signals recorded at M_1 and M_2 exhibit a strong correlation and can be confidently used for the identification and localisation of blockages.

4.3. Sewer blockage identification and localisation

To enhance blockage detection under both clean and blocked pipe conditions, a broadband external excitation source (e.g. white noise generated by a mobile phone in this study) was injected at the upstream end of the pipe (Tank 1). The first spike in the paired-IRF corresponds to the incident wave injected into the pipe. The influence of flow and blockage on the acoustic response in the sewer pipe system is shown in Fig. 5. This figure illustrates how the paired-IRF method distinguishes between a clean and a blocked pipe by comparing the acoustic responses

recorded at two measurement points M_1 and M_2 . Note that each spike in the paired-IRF trace represents a reflection from the pipe system. Therefore, the detection of pulses at times that do not match with the expected arrival times of boundary reflections may indicate the presence of an abnormality in the system. In Fig. 5, all sub-figures exhibit similar features, including a prominent peak at 1.50 ms which corresponds to the propagation delay of the incident wave from M_1 to M_2 , and a pair of spikes due to the reflection of Tank 2 at 62 ms.

In the clean pipe scenario (Fig. 5(a)), the water surface in the pipe is low and stable, so the effective cross-sectional area A remains almost constant. Since the wave speed a and gravitational acceleration g are unchanged, the characteristic impedance $B = a/g A$ remains uniform along the pipe. As a result, the injected white noise travels smoothly, and the paired-IRF response remains flat along the pipe until a clear reflection of the downstream tank (Tank 2). This indicates that the sound waves do not encounter any internal obstructions during their propagation through the pipe and the response mainly represents the effects at the system boundary.

In contrast, when a blockage is present in an empty sewer pipe, it introduces a sudden reduction followed by a subsequent expansion of the effective cross-sectional area (Fig. 5(b)). Once the acoustic wave (white noise) first reaches the front side of the blockage, it encounters a significant change in the characteristic impedance B from the upstream condition (B_1) at the diameters of D_1 , to the blockage point (B_2) at D_2 . At that time $B_1 < B_2$ as a smaller cross-sectional area A leads to a larger impedance at the blockage. This impedance mismatch causes part of the wave to be reflected. The sudden positive spike paired with a negative spike (solid circles in Fig. 5(b)) in the paired IRF corresponds to this upstream reflection. After partially transmitting through the blockage,

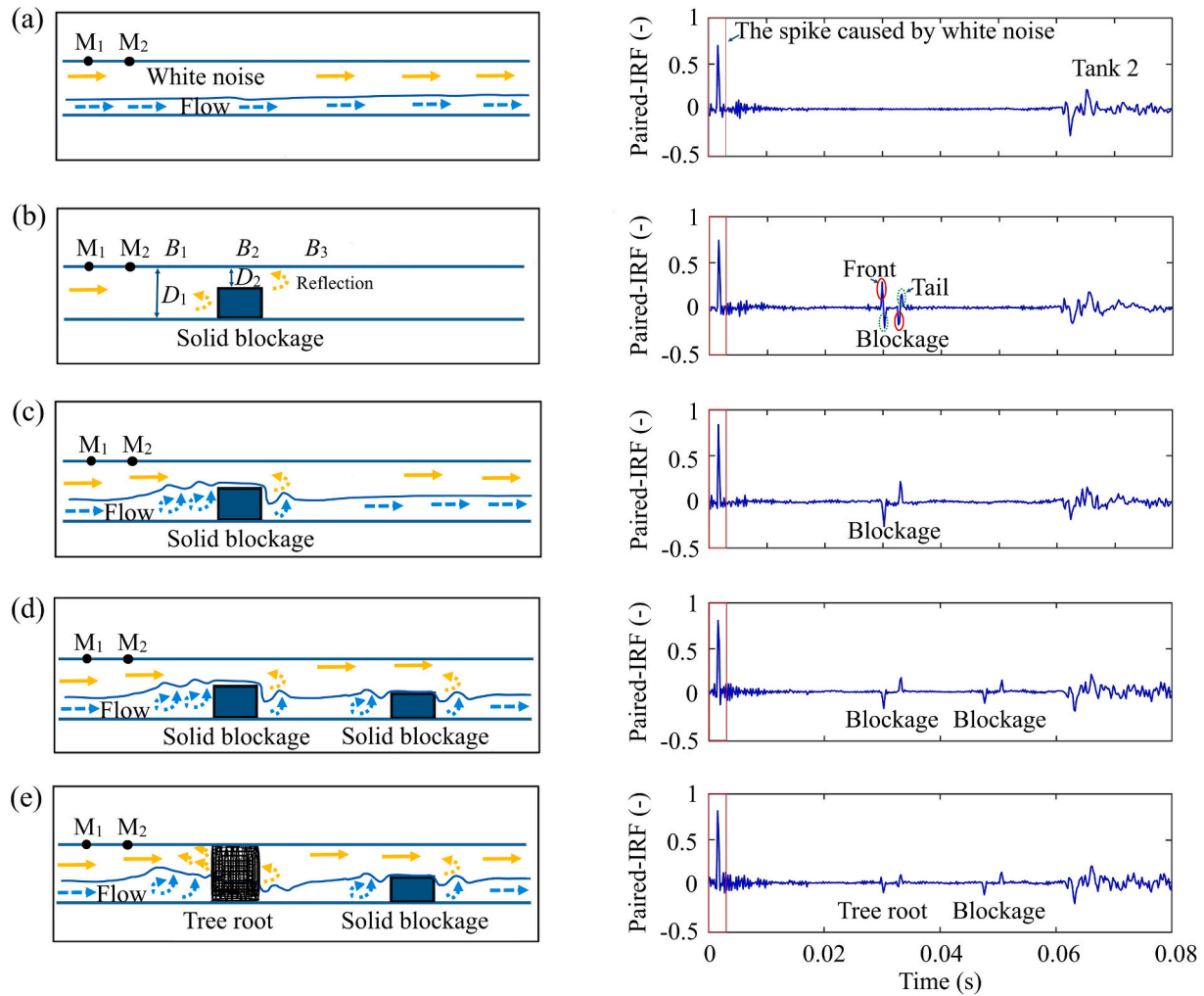


Fig. 5. The different paired impulse responses under flow and blockage conditions. (a) Test 4 – white noise and flow; (b) Test 5 – white noise in blocked pipe without flow; (c) Test 6 white noise, flow and blockage; (d) Test 9- white noise, flow and two solid blockages. (e) Test 9- white noise, flow and two blockages (tree root and solid material).

the wave continues with a smaller effective area and larger B . When the white noise passes completely through the blockage, it transitions to a full pipe with a smaller which equals to B_3 , creating a second reflection in the opposite direction, seen as a negative spike paired with a positive spike (dotted circles in Fig. 5(b)) in the paired IRF. This notes that the reflections (or spikes) in Fig. 5(b) are primarily due to the structural discontinuity introduced by the obstruction. The clarity of this reflection confirms that the paired-IRF method is effective in blockage detection in the dry condition of the pipe. The location of blockages can be calculated by extracting the time of the paired spike. The average time of the red-circle spikes is 31.15 ms and using the speed of sound in air (343 m/s), the distance between the front of the blockage to $M_1 = (343 \times 0.03115) / 2 = 5.34$ m. Similarly, the average time of green-circle spikes is extracted

at 31.35 ms to calculate the distance between the blockage tail to $M_1 = 5.377$ m. Therefore, the estimated thickness of the blockage is 37.7 mm, very close to the actual value of 40 mm.

Under the flow conditions in Fig. 5(c), the blockage causes a gradual rise in upstream water levels, which slowly changes the effective area. However, this change in cross-sectional area A is not sharp enough to create a strong impedance mismatch, so the early reflection is small or absent. The dominant reflection occurs when the wave reaches the tail of the blockage, where the sudden transition from high to low B generates a negative spike paired with a positive spike in the paired IRF. This reflects the partial transmission of the wave through the constriction.

In Fig. 5(c), the blockage reflection is identified with a time delay of approximately 31.55 ms. Therefore, the distance from M_1 to the

Table 2

The accuracy of blockage locations using the paired-IRF method.

Blockage size	Location	τ_1 (ms)	τ_2 (ms)	Average of round-trip travel time $\tau = (\tau_1 + \tau_2)/2$ (ms)	Calculated distance x using (Eq. (4)) (m)	Measured distance from blockage tail to M_1 (m)	Error (%)
Size 1 – Small	BL ₁	30.2	33.1	31.7	5.428	5.380	0.89
Size 2 – Medium	BL ₁	30.1	33.0	31.6	5.411	5.380	0.57
Size 3 – Large	BL ₁	29.9	32.8	31.4	5.377	5.375	0.03
Tree root	BL ₁	30.1	33.1	31.6	5.419	5.380	0.73
Size 1 – Small	BL ₂	47.5	50.5	49.0	8.404	8.380	0.28

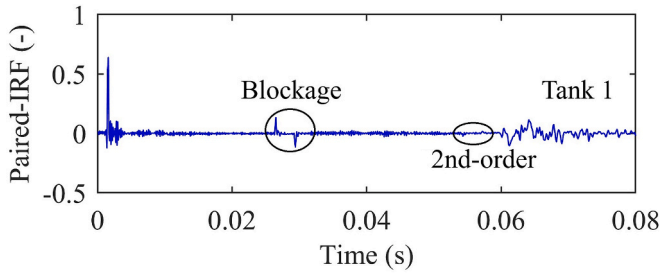


Fig. 6. The paired impulse responses under flow and blockage conditions with sensors placed near the downstream tank.

blockage tail is 5.41 m. This estimation is 3.1 cm longer than the physically measured distance (5.38 m, including blockage thickness of 4 mm), resulting in a low error rate of approximately 0.57%. These results demonstrate that the paired-IRF method is sensitive to both the presence and location of blockages and remains effective under varying hydraulic conditions. Furthermore, the comparison between Fig. 5(b) and 5(c) indicates that while flow conditions can introduce additional background noise, affecting the amplitude and clarity of the blockage signature, they do not prevent the detection of the obstruction. This finding reinforces the robustness and practical applicability of the paired-IRF method for proactive sewer system monitoring under operational flow conditions, as detected anomalies can be integrated with work order systems to enable targeted and efficient inspections.

Fig. 5(d) and 5(e) demonstrate the capacity of the paired-IRF method to detect real situations in sewer systems with multiple blockage events built up by different materials. Once two solid objects (size 2 and 1) were placed in the system (Fig. 5(d)), two pairs of negative and positive spikes appeared in the paired IRF. Compared to a single blockage, two blockages caused more complex reflections, leading to slight noise after the prominent peak. However, the reflected signals still indicated estimated locations of 5.411 m and 8.404 m, respectively, close to the measurement values at BL₁ (5.38 m) and BL₂ (8.38 m) as shown in Table 2. Similarly, signs of two blockages caused by a root and a solid object can be found in the paired IRF in Fig. 5(e). The location of the tree root was also well predicted with an error of 0.73% compared to the actual measurement. The reflected waves due to the tree root were generated from multiple positions along the cross-section area of the tree root

location, making more noise after the prominent peak compared to the response from two solid blockages in Fig. 5(d).

When the sensors were positioned near the downstream tank, the blockage reflection remained clearly identifiable, appearing as a distinct paired spike in Fig. 6. In this case, the paired spikes exhibit an opposite polarity order (positive–negative) compared to upstream placement results (negative–positive) in Fig. 7(a). The appearance of the positive spike occurs because a significant change in characteristic impedance (B) is encountered when the broadband white-noise excitation reaches the front side of the blockage, where the cross-sectional area transitions from large to small. Another noticeable difference in the paired-IRF trace for this setup is the appearance of a small, paired spikes close to the upstream tank, corresponding to the second-order component described in Eq. (1). This observation demonstrates the sensitivity of the proposed method in detecting abnormalities within the sewer system. The second-order component was not observed in upstream sensor placement tests because it was masked by reflections from the tank. Overall, these results demonstrate that the paired-IRF approach is capable of detecting upstream blockages from downstream measurement locations as well.

4.4. The impact of different blockage sizes on blockage detection and localisation

The paired IRF analysis for different blockage sizes under the white noise and flow rate conditions in the pipe is shown in Fig. 7. The corresponding blockage locations (i.e. distance from M₁ to the blockage tail) for each size are presented in Table 2. In Fig. 7(a) the paired IRF displays a small but noticeable paired spike at around 31.65 ms. This signal represents the partial reflection caused by the small obstruction (blockage size 1) in the pipe. The reflected signal is relatively weak but still distinguishable from background noise, demonstrating the ability of the method to detect blockages at a small scale. The estimated distance from M₁ to the blockage tail based on the time delay was 5.428 m, just 4.8 cm different from the measured distance (5.380 m).

Once blockage size 2 appears in the system, the paired-IRF method shows a more prominent paired spike than that of blockage size 1 as shown at 31.55 ms in Fig. 7(b). The increased magnitude suggests that a larger portion of the acoustic energy is being reflected due to a more significant obstruction. This increase in spike magnitude also confirms the sensitivity of the paired-IRF method to the severity of the blockage,

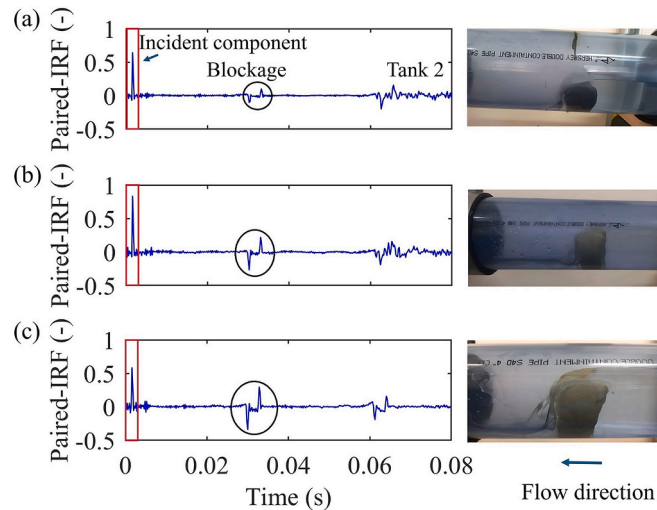


Fig. 7. The paired IRF extracted for three blockage sizes. (a) Blockage size 1 (37% blocked); (b) Blockage size 2 (54% blocked); (c) Blockage size 3 (82% blocked). During the blockage event, the paired-IRF method identified an abnormal reflection signal occurring at approximately 31.6 ms. The larger the blockage, the greater the extracted spike.

Table 3

Estimated blockage area ratios based on reflection ratios received from the paired IRF.

Blockage size	Measured blockage area ratio (%)	Reflection ratio r (-)		Estimated blockage area ratio (%)			Error (%)
		Negative spike	Positive spike	Negative spike	Positive spike	Average	
Small at BL ₁	37	0.24	0.22	38.8	36.5	37.6	1.7
Medium at BL ₁	54	0.45	0.37	62.0	54.4	58.2	7.8
Large at BL ₁	82	0.80	0.71	89.0	82.7	85.8	4.7
Tree root at BL ₁	46	0.28	0.29	44.2	44.3	44.3	-3.8
Small at BL ₂	37	0.23	0.23	37.3	37.7	37.5	1.5

demonstrating its ability not only to detect the presence of blockages but also to characterize their sizes. For blockage size 3 shown in Fig. 7(c), the reflected signal becomes clearer and more distinct. The peak reaches a higher magnitude than those in the previous two cases. This strong reflection demonstrates that the acoustic wave undergoes significant impedance mismatch. The results in all three cases consistently show that as the blockage size increases, the magnitude of the paired spikes in the paired IRF grows proportionally. The detection results for blockage locations in Table 2 also represent that the larger the blockage size, the more accurate the location is. Within the experimental data, the error of the paired IRF in location detection was under 1.0%, specifically 0.89% for small blockages, 0.57% for medium blockages, and -0.03% for large blockages. Accurate location prediction can support predictive maintenance planning, helping water utilities reduce maintenance costs and minimise the impacts of sewer overflows.

4.5. Estimating blockage area ratios

Table 3 presents the estimated blockage area ratios from the reflection coefficients measured by the magnitudes both in the negative and positive spikes in the paired-IRF response. The reflection ratio r was used to compute the blockage size according to Eq. (6), and the resulting estimates were compared against the measured blockage area ratios. For the small blockage (Size 1) with a measured area ratio of 37% at the location BL₁, the reflection ratios from the negative and positive spikes were 0.24 and 0.22, respectively. The estimated blockage area ratios were 38.8% and 36.5%, yielding an average of 37.6%. This value is close to the measured value, showing a minimal error of 1.7%. A small error with 1.5% was also identified for this blockage size at the location BL₂.

For the medium blockage (Size 2), a greater discrepancy between the reflection ratios was observed with values of 0.45 for the negative spike and 0.37 for the positive spike, leading to estimated blockage ratios of 62.0% and 54.4%, respectively. The average of these two estimates provides a blockage size of 58.2%, overestimating the actual blockage (54%) by 7.8%. The relatively larger error using the negative spike is mainly due to some ripples after the major spike in the paired-IRF trace causing the large error when estimating the reflection ratio.

In the case of the large blockage (Size 3), the reflection ratios were 0.80 (negative spike) and 0.71 (positive spike), producing blockage

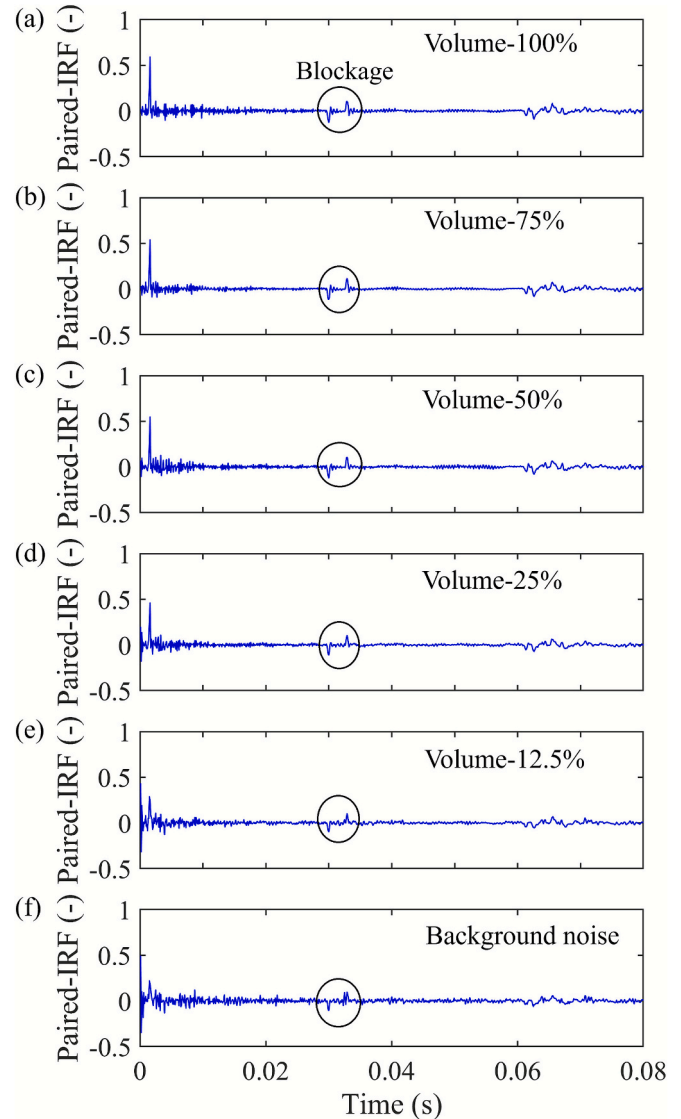


Fig. 9. The influence of the different excitation noise levels on the robustness of the paired IRF. (a) Excitation noise corresponding to the phone volume of 100%, (b) Volume-75%, (c) Volume-50%, (d) Volume-25%, (e) Volume-12.5%, (f) Background noise (no excitation noise).

estimates of 89.0% and 82.7%, respectively. The average estimated ratio was 85.8%, resulting in a moderate error of 4.7%. Unlike overestimating the size of solid blockages, the paired-IRF method underestimated the size of the tree root with an error of -3.8% as the reflection from the root structure was more diffuse and less defined compared to solid objects. Overall, the method performs well for estimating blockage sizes. These results validate the applicability of the paired-IRF-based reflection ratio method for estimating blockage size under flow conditions, with errors remaining under 8% across all cases. In other words, these

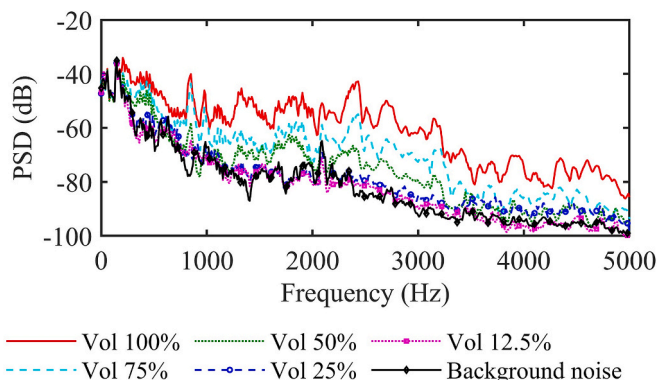


Fig. 8. PSDs of the signals recorded with the different excitation noise levels.

Table 4
Spike amplitudes and SNR values for different excitation volumes.

Excitation Volume (%)	Peak amplitude of main spike-white noise (–)	Average amplitude of paired spike caused by blockage (–)	SNR (dB)
100	0.592	0.115	8.2
75	0.538	0.114	3.6
50	0.545	0.113	1.2
25	0.460	0.107	–0.5
12.5	0.288	0.099	–1.5
BG	0.221	0.099	N/A

findings indicate that the paired-IRF method can effectively identify the presence and estimate the severity of blockages in a sewer pipe by analysing the reflected acoustic signatures. This represents a significant advancement in sewer system monitoring and environmental management. Accurately identifying both the location and severity of blockages enables water utilities to clean blockages proactively before they lead to overflows and thereby reducing environmental impacts and risks to public health.

4.6. The effect of excitation-source volume

To assess robustness under realistic noise conditions, experiments were performed using different excitation-source magnitudes by adjusting the mobile-phone output level (100%, 75%, 50%, 25%, and 12.5%), while keeping the pipe condition (medium blockage) and laboratory environment constant (laboratory pumping). As shown in Fig. 8, the power spectral density (PSD) results confirm that increasing the excitation volume raises the broadband energy across the measurement band. The paired-IRF responses consistently appear at the same time locations and with opposite polarities for all tested volumes (Fig. 9). Both the main injection spike and the blockage-induced paired spikes scale proportionally with excitation energy (Table 4). In Fig. 9, the paired spikes remain observable down to the lowest tested level (12.5%), even for the case of no deliberate excitation (background noise). This demonstrates an important advantage of the proposed method: strong environmental noise travelling through the pipe can act as an effective natural excitation source, reducing the need for externally generated white noise in some real-world conditions. Nevertheless, the calculated SNR values show that detection reliability improves substantially with excitation strength, and an excitation volume above approximately 25% is recommended to ensure consistently robust blockage identification using the paired-IRF method.

5. Conclusion

This study investigated the use of paired acoustic sensors to monitor blockages in sewer pipes. Laboratory experiments were conducted under a variety of controlled conditions to distinguish between clean and

blocked pipe cases. Experimental data were used to evaluate the effectiveness of the paired-IRF technique in blockage detection and localization. The results demonstrated that the presence of blockages produced distinct paired spikes in the paired-IRF output, with the magnitude of the reflected spike increasing proportionally with the blockage size. This confirms the sensitivity of the method not only to the presence but also to the severity of obstructions. The experimental findings validate that the paired-IRF method can consistently detect and locate blockages under both no-flow and flow conditions. Particularly, this method was also able to localise multiple blockages in a sewer pipe and distinguish blockage types such as solid and tree root, with the location errors of less than 1%. These results highlight the practicality of the method for real-world sewer applications. Building on this, the study applied wave reflectometry principles to estimate the blockage size using the reflection ratio calculated from paired-IRF spikes and the impedance changes in the pipe. The findings shown that blockage area ratios could be accurately estimated, with errors below 8.0%. However, the current experiments were conducted in a laboratory setup, where the physical pipe length does not fully replicate the long distances typically found between manholes in sewer networks. Future work should involve field-scale testing to validate the effect of pipe length on blockage localization accuracy and further confirm the method's real-world performance. Moreover, variations in blockage geometry, material composition, and boundary roughness can significantly alter the reflection characteristics, potentially leading to overlapping or ambiguous signatures. Therefore, while the paired-IRF response can indicate the presence of complex or composite blockages, additional information (e.g., from multiple sensing points or complementary diagnostic methods) is required to confidently distinguish between blockage types.

Overall, this work demonstrates that paired acoustic sensors combined with advanced signal processing techniques like the paired IRF provide a rapid, non-invasive, and highly accurate approach for proactive blockage detection in sewer systems. In addition to early detection and localization, the ability to estimate blockage severity supports more informed maintenance decisions, reducing operational costs and mitigating the risk of overflows into residential properties or the environment. Based on the findings of this paper, further laboratory tests and data analysis are underway to investigate a more cost-effective sensor design for sewer blockage detection using permanently installed acoustic sensors. The results will be reported in the future to support the widespread adoption of acoustic monitoring in smart sewer management practices.

Declaration of generative AI and AI-assisted technologies in the writing process

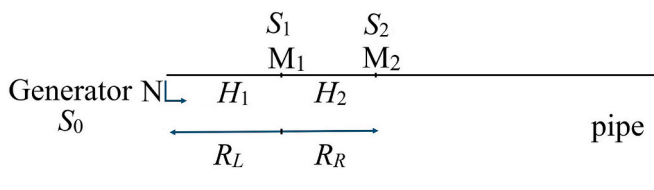
During the preparation of this work the authors used ChatGPT to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Thi Hai Duong Ninh: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Wei Zeng:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Martin Francis Lambert:** Writing – review & editing, Conceptualization. **Nhu Cuong Do:** Writing – review & editing. **Chengcheng Yin:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



S_0 : Injected signal from the acoustic source N
 $S_1(S_2)$: Measured signal at $M_1(M_2)$
 $H_1(H_2)$: Transfer function in N- $M_1(M_1-M_2)$
 $R_L(R_R)$: IRF of the pipe at the left (right) side of M_1

Fig. 10. Schematic diagram illustrating the wave propagation model between the two microphones (M_1 and M_2) used in the paired-IRF formulation.

Appendix A. Theoretical derivation of the paired-IRF equation

In Fig. 10, the generator (N) produces an incident wave that travels through the pipe toward microphones M_1 and M_2 , where H_1 and H_2 describe the corresponding transfer functions. R_L and R_R represent the impulse response function (IRF) of the pipe at the left and right sides of M_1 , respectively.

The total pressures (S_1 and S_2) measured at M_1 and M_2 are composed of the forward- and backward-travelling components:

$$S_1 = S_1^+ + S_1^-, S_2 = S_1^+ H_2 + \frac{S_1^-}{H_2} \quad (A1)$$

The forward and backward pressure waves at M_1 can be expressed in terms of the injected pressure S_0 and the impulse response function at the left (R_L) and right (R_R) sides:

$$S_1^+ = \frac{\left(H_1 + \frac{R_L}{H_1}\right)}{(1 - R_L R_R)} S_0, \quad S_1^- = \frac{R_R \left(H_1 + \frac{R_L}{H_1}\right)}{(1 - R_L R_R)} S_0 \quad (A2)$$

Substituting Eq. (A2) into Eq. (A1) gives:

$$S_1 = \frac{(1 + R_R) \left(H_1 + \frac{R_L}{H_1}\right)}{(1 - R_L R_R)} S_0 \quad (A3)$$

$$S_2 = \frac{\left(H_2 + \frac{R_R}{H_2}\right) \left(H_1 + \frac{R_L}{H_1}\right)}{(1 - R_L R_R)} S_0 \quad (A4)$$

Dividing Eq. (A4) by Eq. (A3) yields:

$$\frac{S_2}{S_1} = \frac{\left(H_2 + \frac{R_R}{H_2}\right)}{(1 + R_R)} \quad (A5)$$

Expanding Eq. (A5) with higher-order wave reflections:

$$\frac{S_2}{S_1} = H_2 + \left(\frac{1}{H_2} - H_2\right) R_R - \left(\frac{1}{H_2} - H_2\right) R_R^2 + \dots \quad (A6)$$

Data availability

Data will be made available on request.

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